

TECHNOLOGY:

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THE HUMANIZATION OF TECHNICS

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I

TECHNICAL MAN is not a contemporary creation. As far as we can go into human history, we see men with various instruments and technics for making things. We see men collaborating or competing with their environment, systematically organizing their activity. It is of note that certain ages of pre-history are naturally known by words which make them successive stages of technical civilization: the stone age, the bronze age, the iron age.

A civilization could hardly be established without considerable use of tools, nor without the help in some de-

gree of different kinds of technical knowledge. But we find all over the world, in ruins or in remains, works of art (in the sense of the architect and the engineer) that only the existence of civilization can explain.

Animal species are adapted to their habitat, or they adapt themselves to given surroundings with a certain stability. But man seems to be endowed by nature with the faculty of modifying his environment, changing at will any surroundings in which he is placed. But we must not think that technical man is artificially grafted onto a natural man from whom he progressively separates himself. Technical man is only one dimension, among other possible aspects, of natural man. We must recognize, nevertheless, that the technical man of our civilization has made great use of his ability to change his environment and the conditions of his

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existence. And the evolution in which his own activity involves him takes on a more and more rapid rhythm — so rapid that he can ask himself if his power to evolve is not also the power to jeopardize himself.

Technical man has always been conscious of his power. It seems that now he is becoming aware of his responsibility. Once experienced, this responsibility appears in most diverse forms, proximate and remote, responsibilities of today and those of tomorrow. The outlines of a problem begin to emerge: the technician asks himself to what extent it is himself, rather than anyone else, who must deal with the problem.

If it were merely question of admiring and of being amazed, technic — especially modern technic — would merit every praise. But no one doubts that there is question of something entirely different. Technic is the work of man, but it produces forces that react upon man. These forces rest upon man more firmly than the thumb of the sculptor on the clay or on the wax which he moulds. Artisan of himself, man changes himself in his customs, his behavior, his tastes, his ideas, his moral . . . Technician without peer, will he be as capable and sure an artisan in remaking his own person? We doubt that he will be, and with reason, it seems.

Can the problem now be sketched in general outlines? Let us open the book of reproaches addressed to modern technic. These reproaches are not all justified, but each of them uncovers an aspect of the problem none the less.

MORAL AND RELIGIOUS REPROACHES

Recent discussions of human relations in terms of industrial efficiency were the occasion of strong protests. The reason for the protests is easily under-

stood: the discussion of human relations could be the occasion of restoring concern for the human person in industrial organization. No matter what is done along this line, if it is of selfish intent, it is doomed to failure. Only a spirit of charity and solidarity can guarantee anything of real value. The same basic demand is presented in many different ways. Whether we speak of the respect or the restoration of the dignity of the human person, of the maintenance or protection of liberty, or of the right of everyone not to be swallowed up in the mob, there is always question of moral values, of rights essential to the human person.

Take, for instance, two inventions, one as wonderful as the other — the telephone and moving pictures. They have, perhaps, the same technical value. But do they have the same human value? The telephone was rapidly integrated into our activities as an indispensable tool. To many of us the movies have become equally indispensable, but not as an instrument for work. They are indispensable as a pleasure which we can no longer give up.

Certainly man is not made for work alone. But he is made even less for the headlong pursuit of pleasure. The quest for his own pleasure can only turn him away from more essential preoccupations. Man should never be too distracted. He will lose the meaning of his life, which is above all the realization of a moral ideal. Pleasure is not wrong. But the abuse or the pursuit of pleasure destroys the appreciation and appeal of an altogether different quest, which alone can satisfy him. Of what value are the gifts and pleasures which technic heaps upon us, if we must pay for them with something which is or ought to be essential for us?

In all the reproaches in this group a serious fear expresses itself, that the evolution which we are undergoing and which we shall continue to undergo in the field of technic threatens the very foundations of our moral being.

REPROACH OF CLASSICAL CULTURE

The idea is quite common today that our culture is threatened, the form of culture which is closely bound up with western civilization and seems inseparable from it. But what is culture? And what is a cultured man? It is hard to define them exactly. We shall be content with sketching some characteristics of culture. As to art, literature, history and the kinds of knowledge which contribute to the formation of a cultured mind, culture takes on two essential aspects: an active aspect, in the creation of artistic, musical and literary values, and the expression of experience having a certain universal value; and a passive aspect in the acceptance and understanding of the values created, and more generally in the interest given to all spiritual conquests.

In its active form, culture comes forth like a spring from the need and desire to express oneself, from the search for proper forms and means of expression and from their use. What, then, is its nature? It is a unique experience, it is an emotional and intellectual experience that admits no substitute. In its passive aspect, culture is the acquisition and refinement of the media and the forms of sensibility that gather in this very special experience. (In its reserve, the expression which we have used is evidently too weak. It would be more accurate to speak of a sensibility capable of receiving the emotional or intellectual message with a pleasure going all the way to ecstasy). In this passive aspect, our culture is bound up

with a certain sensibility or hedonism of an undeniable existential character.

Is this culture threatened by the development of the sciences and of technic? The fact seems incontestable. It is, moreover, inevitable. In the face of the Promethean adventure of technic, the hedonistic base of traditional culture appears to contract and take on a slight nuance of egoism. The popularization of refined and scientific knowledge and the ease with which man can distract himself do not favor the efforts demanded for every refinement of the intelligence or sensibility. The threat is real. The rather classic ideal of culture and of the cultured man appears to be deeply affected by the evolution into which we are drawn by the coordinated progress of science and technic. But must we say that this is a loss which has no compensation?

LIMITS OF HUMAN NATURE

There is a third group of reproaches which cannot be passed over in silence. Their origin lies in a certain notion of the human creature, and the limitations placed on his action and his desires. A certain fear seems to have hovered over the civilizations of the past, a fear of going beyond the rights granted to man in the world, of claiming a knowledge or power which did not belong to him. Such a fear is showing up again in the presence of the constant and unexpected progress of the sciences and technic.

Certainly, to repeat the point, man is not a being predetermined to any particular environment. On the contrary, his key ability appears to be his power to transform his environment and to adapt himself to circumstances which he himself creates. But it would be absurd to attribute to him arbitrary or unlimited power of self-transformation.

tion. His bodily composition is stable, his anatomy is fixed, his physiological functions obey inviolable laws, even his adaptive functions cannot be changed at will.

Technic introduces man's body to artificial surroundings, for which, sometimes, it was not made. In each transformation of his environment man is exposed to risks which he can only know in part. Not all the dangers are evident at the outset. If we indefinitely set aside the more or less natural conditions of our existence, must not the day come when our existence will be hopelessly compromised? The development of scientific knowledge deceives us regarding the suppleness of our mind. Our affective and moral life also have their own structure and are subject to their own laws. We know that this structure and these laws correspond more or less adequately to the common life, to social activity in the human collectivities existing today. But we do not have the assurance that this would be the case in a society which an insufficiently informed and enlightened will would have to organize. The fears occasioned by the rapid development of atomic physics and its applications need be only recalled.

Technic, it is said, changes at an accelerated pace the relationship between man and the whole animal kingdom. But is it absurd to think that something essential in the very being of man is involved in this relationship? If this relationship changes too much or too quickly, can we be sure that we will not suffer the consequences?

CONCLUSION

Let us return to the point where we were before listing all these reproaches. These should help us to distinguish now the contours of the prob-

lem which technicians begin to see. Are we now in a position to draw some conclusions?

Even if these reproaches are not all and completely justified, it would be foolish to cast them aside without examination. Technical man is not the whole man. Each one of the reproaches (and the list is far from exhaustive) is based on an inalienable aspect of man: man in so far as he has a body, man in so far as he has a mind and morality (since he is a person), man in so far as he is a member of society, man in so far as he is an individual in the human species. All these aspects are involved in the ascendancy of technic. What we must take account of is that in each instance we ourselves are not untouched.

II

We cannot be content with merely acknowledging the problem. It is far from being defined with precision. However little we feel our responsibility is involved, our first need is to understand the problem more clearly.

What is to be done? First of all, in the face of the preceding conclusions, a quite pessimistic position suggests itself: technical progress seems to be uncontrollable. However forced this hypothesis may appear, let us suppose that *we* (that is, all those who are associated in technical work) have decided to take control of the process, have decided at least to try to control and direct it.

A preliminary question presents itself. Has anyone heard of an on-going process that can correct itself, reorientate itself without deviating from its fundamental course? We now have many examples of this. The *feed-back* is the means used. So that a *feed-back* work, there must be an examination

of the phenomenon going on. The information resulting from this examination must be incorporated into the conditions determining the phenomenon. And finally, a modification of these conditions in terms of a program to be carried out. Let us use this example as a lead in working out a somewhat bold analogy.

RESEARCH ON MAN AND SOCIETY

First of all, in what does the examination of an on-going phenomenon consist? For the purpose we have in mind, it would not suffice to take into consideration (if that were possible) the state, progress and methods of technic. Our interest must bear also and particularly on the milieu which technics modify — man in his totality and on human societies in all their complexity. How could we take control of our action on man, if man would remain forever “the unknown?” Like the total reality that makes a man, the total reality that makes a society is complex and in a great measure hidden. How can we go about evaluating the action of technical inventions upon a society, if society also remains for us “the unknown?”

What do we wish to say? Simply this: if technicians mean to take their responsibilities seriously in order that the age of technic remain fully human, they cannot dispense with a study, a “complete” study of two complementary things: man and men, societies and civilizations.

We have just alluded to a work of Alexis Carrel, *Man, the Unknown*. We wish in this way to underline the full scope of what we have called the “complete” study of man. Everything in him presents itself as something to be known (the importance of which is itself something to be recognized) from

his physiological being to his moral being, from the structure of his intelligence to that of his affectivity and his impulses. However important it is to know the conditions of his welfare, it is just as important to know the conditions of his happiness. As it is necessary to learn what will guarantee his bodily health, it is also necessary to learn what will guarantee harmony in his psychic life. There is question of an entire program of research to be made by man on man.

We spoke also of societies and civilizations, those “other unknowns.” We cannot indicate even briefly the different kinds of knowledge these unknowns would occasion. This would be a second program of research which man would have to make, if not directly on himself, at least on “the groups” which include him. But the method proper to this sort of research is not given to us ready-made. The corresponding disciplines are scarcely founded. But whoever says “method” also says “theory of knowledge”. (We could illustrate how much the fruitfulness of research can depend upon reflection on the method proper to the research. How to avoid a vicious circle, for example in the analysis of the relationship existing between a person and the environment which contributed to his formation, the environment itself being dependent on the persons who compose it).

Technic aware of its responsibilities, then, cannot dispense with a theory of knowledge upon which its action depends.

That is the first of the three points of our analogy. The second can be reduced to this: that we take into consideration the results of the “complete” study of which we have spoken.

The third point concerns retroaction, based on the observation, which modifies the movement in terms of the effect sought. This third point is twofold. If the retroaction is to be efficacious, the means influencing the phenomenon and its causes must have been attended to. And the goal to be reached must have been proposed.

Let us put the question of means in parentheses. What, then, is the goal in the analogy? There is question, of course, of ends in view of which we have decided (according to our hypothesis) to control the course of technical development. Are these ends inescapable then? Not at all; we chose them ourselves. However powerful technics may be, they cannot accomplish all goals at the same time. For example, do we not have to choose between seeking pleasure and seeking happiness?

This one example will sufficiently clarify our question. What is at stake is judging technical accomplishments and their results while using man as the measure of our judgment. It is the question of giving them a human value, of making them technical and ethical things at the same time. But this is more than just acting or knowing, this is philosophizing above all. No technic, once it is aware of its responsibilities, can get along without a philosophy, unless it is a religion.

The general lines of the problem begin to reveal themselves and, at the

same time, the accompanying difficulties which no radical solution can avoid. Here it is possible to answer the reproaches of those who defend the human person and culture. To accept its full responsibility any technic must join itself to a theory of knowledge, a moral and a philosophy.

We thus begin to understand that a philosophy is not established in a vacuum or in an absolute. It must spring up from the richness of our experience. But experience in the sciences and in technic is an essential part of philosophy. It is unique. There cannot be question, then, of bringing a fully elaborated philosophy into the sciences and into technic. The philosophy which must guide them cannot be made without them, without their active participation. (How? The example of the *feed-back* could again serve as a guide).

Similarly, culture does not have an existence of its own, independent of the milieu which enables it to flourish. We cannot hope to transfer the culture of the past unchanged into the new atmosphere of the age of technical development. But if not all cultural values can be integrally reassumed in the new age why should not new cultural values arise?

The person itself does not have an existence of its own. It is an aspect and expression of the being which we are and which bears the mark of the milieu which contributed to its formation. The technical age will respect the person which the inviolable demands of morality will develop in the new age.

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